



Intelligence. Impact.

Intelligent LLM Decision Engine

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Problem Statement



Current LLM systems lack dynamic decision-making for different queries.
They fail to balance cost, latency, and accuracy effectively

Objectives



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- Build a system that dynamically selects the best strategy per query
- Optimize the trade-off between:
 - Accuracy
 - Latency
 - Cost
- Enable continuous learning from real outcomes
- Improve overall system efficiency over time

Core Idea



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- Focus on improving decision-making, not the model itself
- Predict the best strategy before execution
- Execute only the optimal path
- Learn from outcomes and refine decisions over time

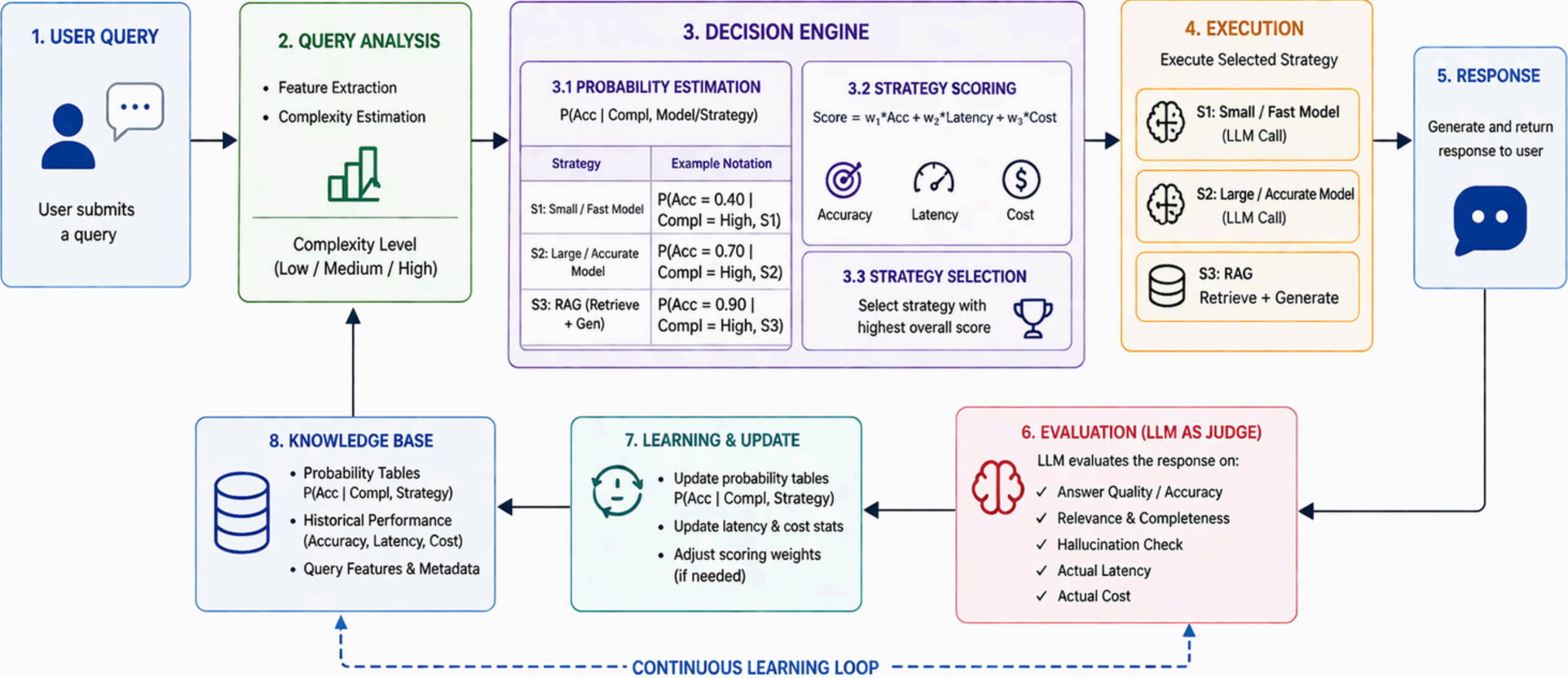
Architecture



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INTELLIGENT LLM DECISION ENGINE

Optimizing Cost, Latency, and Accuracy in Real-Time





Evaluation Metrics

- Accuracy measured using quality score (0 to 1)
- Latency measured as response time
- Cost measured as token or API usage
- Compare baseline system with adaptive system

Expected Results

- Reduced operational cost
- Faster response time
- Maintained or improved accuracy
- Improved decision-making over time



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Challenges

- Complexity estimation may not always be accurate
- Evaluation using LLM-as-judge may introduce noise
- Cold start problem due to lack of initial data
- Building and maintaining RAG knowledge base
- Proper tuning of weights



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Future Work



- Introduce contextual bandit learning for dynamic exploration and exploitation
- Build real-time monitoring dashboard for cost, latency, and performance
- Implement confidence estimation for more reliable decision-making
- Improve evaluation using multi-metric scoring (accuracy, relevance, completeness)

Conclusion



- A decision intelligence layer for efficient LLM usage
- Dynamically balances cost, latency, and accuracy
- Continuously improves through feedback and learning

Thank You